

Unless stated otherwise, all problems are worth 3 points and answers should be stated as exact fractions or decimals rounded to three decimal places. You must show your work to receive credit.

1. A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following (write your answer as a fraction).

a. an ace

c. an ace or a spade

b. the ace of spades

d. A four of spades

2. A coins is flipped and a six-sided die is tossed. Find the following:

a. the sample space

b. the probability of getting a head AND rolling a three.

c. [2 points] the odds of getting a head AND rolling a three.

d. the probability of getting a head OR rolling a three.

e. [2 points] the odds of getting a head OR rolling a three.

3. [6 points each] Complete the following:

- a. The MHU librarian has found that the number of books checked out by students is as follows:

Number of Books	0	1	2	3
proportion	25%	50%	15%	10%

Find the expected number of books checked out by a student.

- b. You have been asked to play a game with a six-sided die. The rules are as follows.

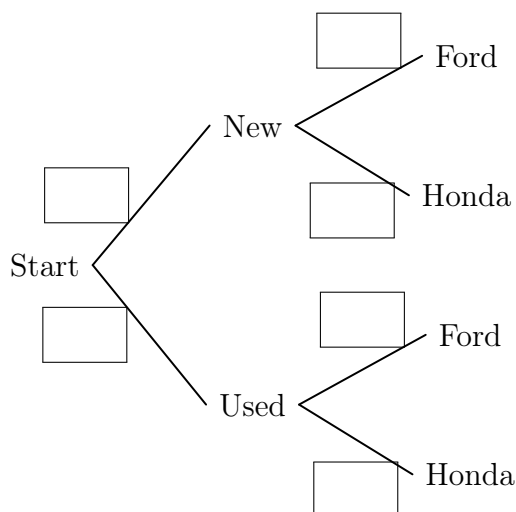
- If you roll a 1 or 2 you win \$50.
- If you roll a 3 you lose \$10.
- If you roll a 4, 5, or 6 you lose \$20

Find the expected value of the game.

4. Approximately 66% of the cars at Big Bill's Cars are new and 34% are used. Of the new cars, 75% are Fords and the rest are Hondas. While 40% of the used cars are Fords and the rest are Hondas.

- a. Complete the tree diagram describing the probability

- b. find the probability that a new car is a Ford.



- c. find the probability that a car is new and a Ford.

- d. find the probability that a car on the

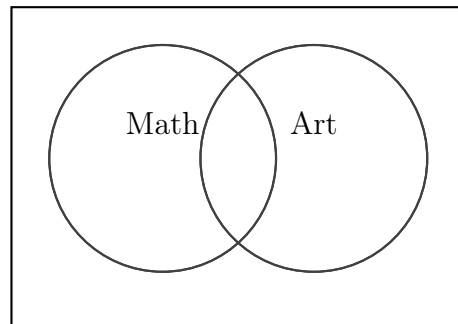
lot is a Ford.

5. Below are the results of a Gallup poll in which workers were asked “Would you prefer a male or female boss?”

	Prefer Male boss	Prefer Female boss	No preference
Male	29%	18%	53%
Female	40%	27%	33%

- a. What is the probability that a worker prefers a female boss?
- b. What is the probability that a worker prefers a male boss given that they are female?
- c. What is the probability that someone is female given that they prefer a male boss?
6. Of the students who are Biology majors, 55% are taking Math, 65% are taking Art, and 30% are taking both courses. Find the probability of the following:

- a. a biology major is not taking Math



- b. a biology major is taking Math or Art

- c. a biology major is not taking Math and not taking Art.

7. A die is rolled. The probability of rolling a 5 is $\frac{1}{6}$, and the probability of rolling a 5 given that a 4 or higher is rolled is $\frac{1}{3}$.
The probability of rolling a 4 or 5 is $\frac{2}{6} = \frac{1}{3}$, and the probability of rolling a 4 or 5 given that an even number is rolled is $\frac{1}{3}$. Justify your answers.

- a. Are the events “rolling a 5” and “rolling a 4 or higher” independent or dependent?
- b. Are the events “rolling a 5” and “rolling a 4 or higher” mutually exclusive or non-mutually exclusive?
- c. Are the events “rolling a 5” and “rolling an even number” mutually exclusive or non-mutually exclusive?
- d. Are the events “rolling a 4 or 5” and “rolling an even number” independent or dependent?

8. [6 points each] A jar contains 30 jellybeans. 12 are red and 18 are blue. Jane randomly picks 5 jellybeans. Find the probability of each of the following. **You only need to write the formulas.**

- a. all 5 are blue.
- b. exactly 4 are blue.
- c. 2 red and 3 blue.
- d. at least 4 are blue.

Honor Statement

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University.

Signature: _____